How well does GPS work?

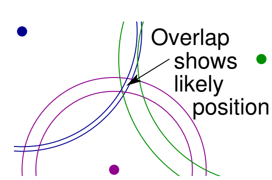
- In the outdoor environment, GPS provides good absolute position measurements anywhere on earth.
- GPS position error depends on many factors, but a well-designed GPS can achieve horizontal accuracy of 3 m and vertical accuracy of 5 m (or better) 95 % of the time.
  - Error is affected by the relative positions of the satellites (want them spread out),
  - Atmospheric refraction (changes speed of propagation of the radio signal),
  - Multipath (bouncing) effects of the received signal,
  - Timing accuracy of the GPS satellites' internal clocks.
- Can be improved—by differential GPS (using a calibrated ground reference signal) and/or satellite-based augmentation systems that can broadcast localized correction factors—to sub-meter accuracies.

Why don't we use GPS indoors?

- In the indoor (or underground) environment, GPS is rarely a good option as an absolute-position sensor.
  - The satellite signal strength is too degraded to be able to synchronize well and get reliable position fixes (no direct view of the satellites).
  - Further, an accuracy of 3 m is usually not good enough for indoor navigation.
- INS-type velocity/acceleration (relative-position) sensors still make sense, but we need an alternative way to GPS to sense absolute position.
- Two common approaches:
  - Trilateration sensors,
  - Triangulation sensors.

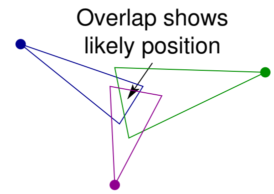
Trilateration

- Trilateration is the process of taking three (noisy) distances to three precisely surveyed points and combining these values to determine a (noisy) estimate of my own position.
  - Distance between "me" and a known point is the same as the distance between that known point and me, so I have drawn arcs from three known points (purple, blue, green), each having some uncertainty band.
  - My position must be within the intersection of these uncertainty bands.
- The distances can be computed based on "time of flight" of a radio signal using ultra-wideband transmitters, or based on received signal strength and a model of how signal strength degrades versus distance.



Triangulation

- Triangulation is the process of taking three (noisy) angles to three precisely surveyed points and combining these values to determine a (noisy) estimate of my own position.
  - I have drawn angles from three known points (purple, blue, green), which have some uncertainty.
  - My position must be within the intersection of the uncertainty bands.
- The angles can be computed using antenna array structures and the MUltiple SIgnal Classification (MUSIC) algorithm, for example (it is based on the measured signal phase differences received by different antennae).
- People sometimes imprecisely refer to trilateration using the term "triangulation," but technically they mean different things.



Beacon methods

- Radio beacons having precisely surveyed self-locations are commonly used with trangulation and trilateration.
  - Ultra-wideband (UWB) radio transmitters can be used to provide distance measures via time-of-flight or received-signal-strength-indication (RSSI).
  - Angles can also be computed to these beacons, for use with triangulation.
- Standard WiFi RSSI, ZigBee, and bluetooth low energy (BLE) have also been reported in the literature.
- While I have suggested that we need three angles or three distances (or some combination of three) to find my own position, we will see later this week that only one angle or distance is truly needed for navigation (with a particle filter, if the navigating platform is in motion).

Non-beacon methods

- Sometimes, we must navigate an environment where it is impractical or impossible to place beacons.
- In this case, we might assume that we know a precise map to the environment.
- Various sensors can provide distances to solid objects in different directions surrounding me; those are correlated to the map to help fix my own position.
  - An example is LIDAR (light detection and ranging), which scans the environment with laser beams and then picks up the reflections.
  - Measuring the time it takes for the light to bounce back makes it possible to judge distance; by scanning, distances to objects in many directions are found quickly.
  - Another example is ultrasound, which transmits a high-frequency sound wave, and measures the time to receive the echo.
  - A third example is RADAR (radio detection and ranging), which transmits a radio-wave frequency and measures the time to receive the reflection.

Summary

- GPS is usually not adequate to provide absolute-position measurements for indoor (or underground) navigation.
  - Signal strength is too poor; accuracy is too low.
- We can still use trilateration and triangulation-type approaches, but not using GPS satellites as the sources.
- Trilateration works based on estimating the distances to precisely surveyed points; triangulation works based on estimating the angles to precisely surveyed points.
- We can use either approach with beacons or with a mapped environment.
  - Computing the correspondence between measurements and the mapped environment to compute position is difficult, but necessary if beacons are impossible or impractical for an application.
  - Very advanced methods also “learn” a map of the environment; e.g., “simultaneous localization and mapping” or SLAM.